

Lecture 01

1. Compare and contrast the roles of an Intermediate System (IS) at the Data Link, Network, and Transport layers. Provide one practical example of each type. (4 marks)

->> An **Intermediate System (IS)** is a device that forwards data between end systems, and its role varies by OSI layer.

At the **Data Link layer (Layer 2)**, an IS forwards **frames using MAC addresses** and works within the same local network.

Example: An **Ethernet switch** forwarding data between computers in a LAN.

At the **Network layer (Layer 3)**, an IS routes **packets using IP addresses** and selects the best path between different networks.

Example: A **router** forwarding packets from a local network to the Internet.

At the **Transport layer (Layer 4)**, an IS examines **port numbers and transport protocols** to manage traffic, security, or load balancing.

Example: A **Layer 4 load balancer** distributing TCP connections among multiple servers.

2. Explain how the end-to-end principle and the “IP over everything” principle together form the foundation of the Internet’s design. Give a real-world example where these principles allow for easy adoption of new network technologies. (4 marks)

->> The end-to-end principle states that intelligence and control (such as reliability, error control, and security) should be implemented at the end systems rather than inside the network. The core network remains simple and only forwards packets, while applications handle functions like retransmission and encryption.

The “IP over everything” principle means that IP acts as a universal layer that can run over any underlying physical or data-link technology (Ethernet, Wi-Fi, fiber, 4G/5G, satellite). As long as a network can carry IP packets, it can connect to the Internet.

Together, these principles form the foundation of the Internet by keeping the network simple, scalable, and flexible. The network does not need to change when new applications or access technologies are introduced; only the end hosts or access layer need updates.

Real-world example:

When mobile networks (4G/5G) were introduced, they easily connected to the Internet because they carry IP packets. Applications like web browsing, WhatsApp, and video streaming worked

immediately without changes to the Internet core, since reliability and control are handled at the end systems using TCP/UDP.

3. Analyze the differences between the OSI model and the TCP/IP model in terms of layer functionality. Specifically, discuss why the Presentation and Session layers are absent in the TCP/IP model and where their functions are implemented instead. (4 marks)

->> The OSI model has 7 layers, while the TCP/IP model has 4 layers. In the OSI model, the Presentation layer handles data formatting, encryption, and compression, and the Session layer manages session establishment, synchronization, and termination.

These two layers are absent in the TCP/IP model because TCP/IP follows a simpler, practical design where such functions are not kept as separate layers. Instead, their responsibilities are merged into other layers.

The Presentation layer functions (encryption, data representation, compression) are implemented in the Application layer of TCP/IP (e.g., HTTPS, TLS). The Session layer functions (session control, synchronization) are mainly handled by the Application and Transport layers (e.g., TCP session management).

Thus, TCP/IP integrates OSI's Presentation and Session layer functionalities rather than defining them separately.

4. Describe how protocol encapsulation and multiplexing/demultiplexing work together in the TCP/IP model. Use the example of an HTTP request being sent over Ethernet to illustrate the process. (4 marks)

->> In the TCP/IP model, encapsulation is the process of adding protocol headers as data moves down the layers, while multiplexing/demultiplexing allows multiple applications and protocols to share the same network.

When a user sends an HTTP request, the Application layer creates the HTTP message.

At the Transport layer, TCP encapsulates the HTTP data by adding a TCP header with source and destination port numbers (multiplexing different applications).

At the Internet layer, IP adds an IP header containing source and destination IP addresses.

At the Network Access layer, Ethernet encapsulates the packet into a frame with MAC addresses.

At the receiver, demultiplexing occurs in reverse: Ethernet passes the data to IP, IP passes it to TCP, and TCP uses the port number to deliver the data to the correct HTTP application.

5. Discuss the hierarchical structure of Internet network organization, from continental backbone networks down to individual hosts. Explain how this hierarchy supports scalability and interoperability across diverse network technologies. (4 marks)

->> The Internet is organized in a hierarchical structure to efficiently connect billions of devices worldwide. At the top are continental and global backbone networks, operated by Tier-1 ISPs, which provide high-speed long-distance connectivity. Below them are regional and national ISPs (Tier-2) that connect cities and countries. Local ISPs (Tier-3) connect homes, campuses, and organizations.

Within local networks, access networks (Ethernet, Wi-Fi, fiber, 4G/5G) connect individual hosts such as computers and smartphones. Routers at each level forward traffic between networks using IP.

This hierarchy supports scalability by limiting routing information and traffic management to appropriate levels, preventing the Internet from becoming unmanageable. It ensures interoperability because all layers use IP as a common protocol, allowing diverse technologies to interconnect seamlessly.

Lecture 2

1. Explain how the IPv4 datagram header fields support fragmentation and reassembly across networks with different MTUs. Use a specific example involving Ethernet and X.25 networks to illustrate your explanation. (4 marks)

->> IPv4 supports fragmentation when a packet travels across networks with different MTUs.

If a packet size exceeds the MTU of the next network, it is fragmented. The following IPv4 header fields help in fragmentation and reassembly:

1. **Total Length** – Indicates the full size of the packet. If greater than MTU, fragmentation occurs.
2. **Identification** – Same value in all fragments; helps the destination group fragments correctly.
3. **Flags** –
 - **DF (Don't Fragment)**: If set, packet cannot be fragmented.
 - **MF (More Fragments)**: Set to 1 in all fragments except the last one.

4. **Fragment Offset** – Indicates the position of fragment data in the original packet (in units of 8 bytes).

Example:

Ethernet MTU = 1500 bytes, X.25 MTU $\approx$ 576 bytes.

A 1500-byte packet from Ethernet entering X.25 is divided into smaller fragments (each $\leq$ 576 bytes). All fragments carry the same Identification number, proper Fragment Offset, and MF flag (last fragment has MF = 0).

At the destination, fragments are reassembled using Identification and Fragment Offset fields.

Thus, IPv4 header fields ensure correct fragmentation and reassembly across networks with different MTUs.

2. Compare and contrast classful and classless addressing in IPv4. Discuss how CIDR alleviates the problem of address exhaustion and routing table explosion, providing an example with a given IP block. (4 marks)

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Classful Addressing:

- IP divided into fixed classes (A, B, C).
- Fixed subnet masks (/8, /16, /24).
- Causes address wastage.

Classless Addressing (CIDR):

- No fixed classes.
- Uses variable-length prefix (/n).
- Efficient address allocation.

Address Exhaustion:

Example: If 2000 IPs are needed, classful system gives a full Class B (65,536 IPs) $\rightarrow$ wastage.

With CIDR, block like **192.168.0.0/21** (2048 IPs) can be assigned $\rightarrow$ efficient use.

Routing Table Explosion:

CIDR supports route aggregation.

Example:

192.168.0.0/24 to 192.168.3.0/24

can be summarized as **192.168.0.0/22** → reduces routing entries.

Thus, CIDR solves address wastage and reduces routing table size.

3. Describe the role of subnet masks in both subnetting and supernetting. Using the example of IP address 141.14.72.24 with subnet mask 255.255.192.0, show how the network and subnet addresses are derived and explain the impact on routing. (4 marks)

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◇ Role of Subnet Mask

A **subnet mask** separates the **network part** and **host part** of an IP address.

- **In Subnetting:**
The mask is extended (more 1s) to create smaller sub-networks within a network.
- **In Supernetting:**
The mask is shortened (fewer 1s) to combine multiple networks into one larger network (CIDR).

◇ Given:

IP Address: **141.14.72.24**

Subnet Mask: **255.255.192.0**

Convert mask to prefix:

255.255.192.0

= 11111111.11111111.11000000.00000000

= **/18**

◇ Find Network Address

Perform bitwise AND:

Third octet block size = $256 - 192 = 64$

Subnets in 3rd octet:

0, 64, 128, 192

Given IP → 72 lies between 64 and 127

So, **Network Address = 141.14.64.0**

◇ Subnet Address

Since default Class B mask is /16,
Using /18 means subnetting is done.

So subnet: **141.14.64.0/18**

Range:

141.14.64.0 → 141.14.127.255

◇ Impact on Routing

- Routers use the subnet mask to determine the correct network.
- Subnetting reduces broadcast domain size.
- Improves routing efficiency and address utilization.

Thus, subnet masks help identify networks in both subnetting and supernetting and improve routing efficiency.

4. Analyze the function of ICMPv4 in the IP network layer. Explain with examples how ICMP error-reporting messages (such as Destination Unreachable and Time Exceeded) assist in network diagnostics and troubleshooting.

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◇ Function of ICMPv4

ICMPv4 (Internet Control Message Protocol) works at the **network layer** and is used for:

- Error reporting
- Network diagnostics
- Control messages

It does not carry user data, but reports problems related to IP packet delivery.

◇ ICMP Error-Reporting Messages

1. Destination Unreachable

- Sent when a packet cannot reach the destination.
- Reasons:
 - Network unreachable
 - Host unreachable
 - Port unreachable

Example:

If a router cannot find a route to a destination network, it sends a *Destination Unreachable* message to the sender.

2. Time Exceeded

- Occurs when the **TTL (Time To Live)** becomes 0.
- Prevents packets from looping forever.

Example:

In the traceroute command, routers send *Time Exceeded* messages when TTL expires, helping to trace the path of packets.

◇ Role in Troubleshooting

- Helps detect routing problems.
- Identifies unreachable hosts or networks.
- Assists tools like ping and traceroute in diagnosing connectivity issues.

Thus, ICMPv4 supports IP by reporting errors and helping in network diagnostics and troubleshooting.

Lecture 11

1. Compare the noise resistance and bandwidth capabilities of Twisted Pair (UTP/STP), Coaxial Cable, and Fiber Optic Cable. Explain how the physical construction of each contributes to its performance in real-world networking environments. (4 marks)

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Twisted Pair (UTP/STP):

Twisted pair cables have low to moderate noise resistance. Twisting of wires helps reduce electromagnetic interference, while STP provides better protection due to shielding. They support moderate bandwidth and are commonly used in LANs.

Coaxial Cable:

Coaxial cables offer higher noise resistance than twisted pair because of their metallic shielding. They support higher bandwidth and are suitable for cable TV and broadband networks.

Fiber Optic Cable:

Fiber optic cables have excellent noise resistance as they use light instead of electrical signals, making them immune to electromagnetic interference. They support very high bandwidth and are ideal for high-speed and long-distance communication.

2. Explain the principle of data transmission in Fiber Optic cables, including the roles of core, cladding, and light refraction. Differentiate between single-mode and multi-mode fiber optics in terms of application scenarios and performance. (4 marks)

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Principle of Data Transmission in Fiber Optic Cable:

Fiber optic cables transmit data in the form of light pulses. The core carries the light signals, while the cladding surrounds the core and has a lower refractive index. Due to total internal reflection, light continuously reflects inside the core, allowing data to travel long distances with minimal loss.

Single-Mode vs Multi-Mode Fiber:

Single-mode fiber uses a very small core and allows only one light path, providing higher bandwidth and longer distance, mainly used in long-distance and backbone networks.

Multi-mode fiber has a larger core that allows multiple light paths, causing more signal loss, and is used for short-distance applications such as LANs.

3. Analyze the use of Power Line Communication (PLC) as a transmission medium. Discuss its advantages and limitations compared to traditional wired media such as twisted pair and coaxial cable, especially in terms of deployment, data rate, and duplex mode. (4 marks)

->> Power Line Communication (PLC):

Power Line Communication (PLC) transmits data over existing electrical power lines, eliminating the need for separate communication cables.

Advantages:

PLC offers easy and low-cost deployment since it uses already installed power lines. It is useful in areas where installing new cables is difficult.

Limitations (Compared to Twisted Pair & Coaxial):

PLC has lower and unstable data rates due to electrical noise on power lines. It is more prone to interference than twisted pair and coaxial cables. PLC generally supports half-duplex communication, whereas twisted pair and coaxial cables commonly support full-duplex transmission.

Conclusion:

While PLC is convenient for deployment, traditional wired media provide higher data rates, better reliability, and full-duplex performance.

4. Discuss why magnetic storage media are still used for large-scale data transfer despite the availability of high-speed internet. Relate your answer to concepts such as bandwidth, cost, feasibility, and security in enterprise environments. (4 marks)

->> Magnetic storage media (such as hard disks and magnetic tapes) are still used for large-scale data transfer despite high-speed internet availability.

First, magnetic media offer very high effective bandwidth for massive data volumes, as physically transporting storage devices (data shipping) can be faster than transferring terabytes of data over networks.

Second, they are cost-effective, providing low cost per gigabyte compared to long-term high-speed network usage.

Third, in many enterprise environments, transferring extremely large data over the internet is not feasible due to bandwidth limits, latency, and downtime risks.

Finally, magnetic storage allows better security, since data can be kept offline, reducing exposure to cyberattacks during transfer.

1. Explain how the twisted design of UTP and STP cables reduces electromagnetic interference (EMI) and crosstalk. Compare their effectiveness and typical use cases in modern network installations. (4 marks)

->> Twisted Pair Design and Interference Reduction:

In UTP and STP cables, the pairs of wires are twisted together, which helps cancel out electromagnetic interference (EMI) and reduces crosstalk by balancing the induced noise in both wires.

UTP vs STP Effectiveness:

UTP relies only on twisting, providing moderate protection and is sufficient for most office and home networks.

STP includes an additional metallic shield, offering better EMI protection, making it more effective in electrically noisy environments.

Use Cases:

UTP is widely used in LANs due to its low cost and easy installation.

STP is used in industrial areas, data centers, and high-interference environments where better noise protection is required.

2. Describe the structural composition of a coaxial cable and explain why it supports higher bandwidth and better noise immunity compared to twisted pair cables. Provide examples of where coaxial cables are still preferred today. (4 marks)

->> Structure of Coaxial Cable:

A coaxial cable consists of a central copper conductor, surrounded by a dielectric insulator, a metallic braided or foil shield, and an outer protective jacket.

Bandwidth and Noise Immunity:

The shielded structure protects signals from electromagnetic interference, providing better noise immunity than twisted pair cables. This design allows coaxial cables to support higher bandwidth and more stable signal transmission.

Current Applications:

Coaxial cables are still preferred in cable television (CATV), broadband internet, and CCTV systems due to their reliability and resistance to interference.

3. Analyze Power Line Communication (PLC) as an alternative data transmission medium. Discuss its operational principles, advantages in terms of infrastructure reuse, and limitations related to data rate and duplex communication. (4 marks)

->> Power Line Communication (PLC) – Principle:

PLC transmits data by superimposing high-frequency data signals onto existing electrical power lines, allowing the same cables to carry both power and data.

Advantages:

Its main advantage is infrastructure reuse, as no additional cabling is required, making deployment cost-effective and easy, especially in buildings where new wiring is difficult.

Limitations:

Power lines are not designed for data transmission, so PLC suffers from high noise and interference, resulting in lower and unstable data rates compared to twisted pair or coaxial cables.

PLC generally supports half-duplex communication, limiting simultaneous two-way data transfer.

Conclusion:

PLC is convenient for quick deployment but offers lower performance and reliability than traditional wired media.

4. Compare single-mode and multi-mode fiber optic cables in terms of light transmission, distance capabilities, and typical applications. Explain why single-mode fiber is preferred for long-distance communication. (4 marks)

->> Single-mode vs Multi-mode Fiber Optic Cables:

Single-mode fiber transmits light through a very small core, allowing only one light path. This results in minimal signal loss, enabling long-distance communication and very high bandwidth. It is mainly used in telecommunication backbones, WANs, and long-haul networks.

Multi-mode fiber has a larger core that allows multiple light paths, causing more dispersion and limiting the transmission distance. It is commonly used in short-distance applications such as LANs and data centers.

Reason for Preference of Single-mode Fiber:

Single-mode fiber is preferred for long distances because it has lower attenuation and no modal dispersion, providing better performance over large distances.

5. Evaluate the scenarios where physical magnetic media (e.g., tapes, disks) are still used for data transfer despite the availability of high-speed networks. Consider factors such as data volume, security, cost, and network limitations. (4 marks)

->> Use of Physical Magnetic Media for Data Transfer:

Physical magnetic media such as hard disks and magnetic tapes are still used when transferring very large volumes of data, as physically moving storage devices can be faster than sending terabytes of data over networks.

They are cost-effective, offering a low cost per gigabyte compared to long-term use of high-speed network bandwidth.

Magnetic media also provide better security, since data can be kept offline, reducing the risk of cyberattacks during transfer.

Additionally, network limitations such as bandwidth constraints, high latency, or unreliable connectivity make physical media more practical in enterprise environments.

Lecture 14

1. Compare Circuit Switching and Packet Switching in terms of resource allocation, delay characteristics, and suitability for different types of applications (e.g., voice vs. data). Provide an example where each method is optimally used. (4 marks)

->> Circuit Switching vs Packet Switching:

Feature	Circuit Switching	Packet Switching
Resource Allocation	Dedicated path reserved for the entire session	Shares network resources; no dedicated path
Delay Characteristics	Low and constant delay once connection is established	Variable delay due to routing and congestion
Application Suitability	Best for real-time applications like voice and video calls	Best for data applications like emails, web browsing, file transfer
Example	Traditional telephone networks	Internet, LAN/WAN data networks

Conclusion:

Circuit switching ensures reliable, consistent performance for continuous streams, while packet switching is efficient and flexible for bursty data traffic.

2. Explain why Message Switching is no longer widely used in modern networks. Discuss its limitations in terms of storage requirements, latency, and suitability for real-time communication compared to Packet Switching. (4 marks)

->> Why Message Switching Is No Longer Used:

Message switching stores the entire message at each intermediate node before forwarding it to the next node. This requires large storage capacity at every switch, making it inefficient for modern high-speed networks.

Because the full message must be received before forwarding, it introduces high latency, especially for long messages. This makes it unsuitable for real-time communication such as voice or video, where low delay is critical.

Compared to message switching, packet switching divides data into smaller packets that are forwarded immediately, resulting in lower delay, less storage requirement, and better support for real-time applications.

Conclusion:

Due to high storage needs, large delays, and poor real-time performance, message switching has been replaced by packet switching in modern networks.

3. Describe the three phases of Circuit Switching and explain how this approach ensures reliable communication for voice applications. Contrast this with the connectionless nature of Packet Switching in the Internet. (4 marks)

->> Phases of Circuit Switching:

1. Connection Setup: A dedicated communication path is established between sender and receiver before data transfer begins.
2. Data Transfer: Voice data is transmitted continuously over the reserved circuit with constant bandwidth and fixed delay.
3. Connection Teardown: After communication ends, the circuit is released for other users.

Reliability for Voice Applications:

Because the path and resources are dedicated, circuit switching provides low, predictable delay and no packet loss, making it highly reliable for voice communication.

Contrast with Packet Switching:

Packet switching (used in the Internet) is connectionless, where data is sent in independent packets without a dedicated path, leading to variable delay and possible packet loss, but better efficiency for data traffic.

4. Analyze how Packet Switching improves line efficiency through multiplexing and prioritization. Discuss how Quality of Service (QoS) can be implemented in a packet-switched network using priority-based forwarding. (4 marks)

->> Packet Switching and Line Efficiency:

Packet switching improves line efficiency by using statistical multiplexing, where multiple data streams share the same communication link. The line is used only when packets are transmitted, reducing idle time compared to dedicated circuits.

Prioritization and QoS:

Quality of Service (QoS) in packet-switched networks is implemented using priority-based forwarding, where packets are classified and queued based on importance. Real-time traffic such as voice and video is given higher priority than normal data packets, ensuring lower delay and reduced packet loss.

Conclusion:

Through multiplexing and prioritization, packet switching efficiently uses network resources while maintaining acceptable performance for time-sensitive applications.

5. Compare connectionless and connection-oriented switching approaches in the context of Packet Switching. Explain the role of forwarding tables and pre-established circuits, and discuss the trade-offs in terms of setup delay and reliability. (4 marks)

->> Connectionless vs Connection-Oriented Packet Switching:

In connectionless packet switching (e.g., Internet/IP), packets are sent independently without a pre-established path. Each router uses a forwarding table to decide the next hop for every packet. This approach has no setup delay and is flexible, but packets may arrive out of order or be lost, reducing reliability.

In connection-oriented packet switching (e.g., virtual circuits), a pre-established logical path is set up before data transmission. All packets follow the same route, providing more reliable and ordered delivery, but introducing setup delay before communication starts.

Trade-off:

Connectionless switching favors speed and flexibility, while connection-oriented switching offers better reliability and predictable performance at the cost of setup time.

Lecture 16

1. Compare the Stop-and-Wait and Sliding Window flow control mechanisms in terms of efficiency, resource utilization, and suitability for different network conditions. Provide a scenario where each method would be preferred. (4 marks)

->> Stop-and-Wait vs Sliding Window Flow Control:

Stop-and-Wait:

In Stop-and-Wait, the sender transmits one frame at a time and waits for an acknowledgment before sending the next. This leads to low efficiency and poor link utilization, especially on high-latency networks. It is best suited for simple or low-speed networks with short distances.

Sliding Window:

Sliding Window allows multiple frames to be sent before receiving acknowledgments, improving throughput and resource utilization. It is efficient in high-speed or long-delay networks.

Preferred Scenarios:

Stop-and-Wait is preferred in small, low-latency systems.

Sliding Window is preferred in modern high-bandwidth networks such as the Internet.

2. Explain how Automatic Repeat Request (ARQ) mechanisms handle error control in the data-link layer. Differentiate between Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ in terms of retransmission strategy and buffer requirements. (4 marks)

->> Automatic Repeat Request (ARQ):

ARQ is an error control mechanism in the data-link layer where the sender retransmits frames when errors are detected using acknowledgments (ACKs) and timeouts.

Types of ARQ:

- Stop-and-Wait ARQ: The sender transmits one frame and waits for an ACK. On error or timeout, only that frame is retransmitted. It requires minimal buffering but has low efficiency.
- Go-Back-N ARQ: The sender can transmit multiple frames. If an error occurs, the erroneous frame and all subsequent frames are retransmitted. It needs buffering only at the sender.
- Selective Repeat ARQ: Only the incorrect or lost frames are retransmitted. It requires buffers at both sender and receiver, but provides the highest efficiency.

Conclusion:

Selective Repeat is the most efficient but complex, while Stop-and-Wait is the simplest.

3. Describe the role of positive ACK, negative ACK (NACK), and timeout in error control protocols. Explain how these mechanisms work together to ensure reliable data transmission over an unreliable medium. (4 marks)

->> Role of ACK, NACK, and Timeout in Error Control:

A positive acknowledgment (ACK) is sent by the receiver to confirm the successful receipt of a data frame.

A negative acknowledgment (NACK) is sent when a frame is received with errors, requesting immediate retransmission.

A timeout occurs when the sender does not receive an ACK within a specified time, indicating that the frame or ACK was lost. The sender then retransmits the frame.

Reliable Transmission:

Together, ACK confirms correct delivery, NACK signals detected errors, and timeout handles lost frames or acknowledgments, ensuring reliable data transfer over unreliable channels.

4. Analyze the trade-offs between Go-Back-N ARQ and Selective Repeat ARQ in terms of bandwidth efficiency, complexity, and memory usage at the receiver. In which network conditions would Selective Repeat ARQ be more advantageous? (4 marks)

->> Go-Back-N vs Selective Repeat ARQ:

Go-Back-N ARQ retransmits the erroneous frame and all subsequent frames, leading to lower bandwidth efficiency but simpler implementation. It requires minimal receiver memory, as only in-order frames are accepted.

Selective Repeat ARQ retransmits only the lost or corrupted frames, giving higher bandwidth efficiency. However, it is more complex and requires larger receiver buffers to store out-of-order frames.

When Selective Repeat Is Better:

Selective Repeat ARQ is more advantageous in high-bandwidth, high-latency networks where retransmitting multiple frames would waste significant bandwidth.

5. Illustrate with a diagram or example how the Sliding Window protocol can be combined with ARQ to achieve both flow control and error control simultaneously. Explain how window size affects performance. (4 marks)

->> Sliding Window with ARQ (Flow + Error Control):

The Sliding Window protocol allows a sender to transmit multiple frames (up to the window size) without waiting for individual ACKs, providing flow control. When combined with ARQ, it also ensures error control by retransmitting lost or corrupted frames using ACKs, NACKs, or timeouts.

Example / Diagram:

Sender Window: [1][2][3][4] → sent

Receiver ACKs: ACK 1, ACK 2

Error in Frame 3 → retransmit Frame 3 (ARQ)

Window slides → [5][6] sent

How Window Size Affects Performance:

- Small window size: Lower throughput, better for low-speed or simple networks.
- Large window size: Higher throughput and better link utilization, but needs more buffering and control overhead.

Conclusion:

Sliding Window with ARQ efficiently combines continuous data flow and reliable transmission, making it suitable for modern networks.

Lecture 17

1. Compare and contrast Single-bit, Multiple-bit, and Burst errors in terms of their causes, detection difficulty, and impact on data integrity. Provide an example scenario where each type might occur in a communication system. (4 marks)

->> Single-bit Error:

A single-bit error occurs when only one bit in a data unit is changed, usually due to minor noise or interference. It is easy to detect using parity checks and has limited impact on data integrity.

Example: Electrical noise affecting a single bit in a wired LAN.

Multiple-bit Error:

Multiple-bit errors affect more than one bit, but not necessarily consecutive bits. They are caused by strong interference or hardware faults and are harder to detect than single-bit errors. They have a greater impact on data accuracy.

Example: Faulty network interface card causing several bit errors.

Burst Error:

A burst error affects a sequence of consecutive bits, usually due to severe noise, signal fading, or synchronization loss. Burst errors are most difficult to handle and can seriously damage data integrity.

Example: Wireless signal interference causing corruption of several consecutive bits.

2. Explain how Parity Check works for error detection, including even and odd parity schemes. Discuss its limitations and explain why it is ineffective for detecting multiple-bit or burst errors. (4 marks)

->> Parity Check for Error Detection:

Parity check adds an extra parity bit to a data unit to make the number of 1s either even (even parity) or odd (odd parity).

- Even parity: Parity bit set so total 1s in the data + parity bit is even.
- Odd parity: Parity bit set so total 1s is odd.

Limitations:

Parity check can detect single-bit errors but fails for multiple-bit errors (an even number of flipped bits cancels out parity) and is ineffective for burst errors, where several consecutive bits may change simultaneously.

Conclusion:

Parity is simple and low-cost but only suitable for basic error detection.

3. Describe the Cyclic Redundancy Check (CRC) method for error detection. Explain the role of the polynomial divisor, codeword formation, and the division process at both sender and receiver. Why is CRC more robust than parity check? (4 marks)

->> Cyclic Redundancy Check (CRC) for Error Detection:

CRC is an error detection method that treats the data as a binary number and divides it by a predetermined polynomial divisor.

Process:

1. Codeword Formation (Sender):

- Append $n-1$ zeros to the data (where n = degree of the divisor).
- Divide the modified data by the polynomial using binary division.

- Append the remainder to the original data to form the codeword.

2. Error Checking (Receiver):

- Receiver divides the received codeword by the same polynomial.
- If the remainder is zero, no error is detected; otherwise, an error occurred.

Why CRC is More Robust:

- CRC can detect single-bit, multiple-bit, and burst errors, unlike parity check, which only detects single-bit errors.
- It is widely used in LANs, WANs, and storage devices due to high reliability.

4. Differentiate between Backward Error Correction (BEC) and Forward Error Correction (FEC). Discuss the scenarios where each method is preferred, considering factors such as transmission medium, cost, and latency. (4 marks)

->> Backward Error Correction (BEC):

- In BEC, the receiver detects errors and requests the sender to retransmit the erroneous data.
- It is suitable for reliable, low-latency networks with low error rates (e.g., wired LANs).
- Advantages: Simple, less redundant data transmitted.
- Disadvantages: Retransmission increases delay, not ideal for real-time applications.

Forward Error Correction (FEC):

- In FEC, the sender adds redundant bits so the receiver can detect and correct errors without retransmission.
- Preferred in high-error or long-distance links (e.g., satellite, wireless) where retransmission is costly or impossible.
- Advantages: No retransmission needed, suitable for real-time communication.
- Disadvantages: Higher bandwidth and processing cost due to redundancy.

Conclusion:

BEC saves bandwidth but increases latency, while FEC increases efficiency and reliability in unreliable or high-latency channels.

5. Using the formula $2^r \geq m+r+1$, calculate the minimum number of redundant bits r required for error correction of a 7-bit ASCII character. Explain the logic behind this inequality in the context of Hamming distance and error correction capability. (4 marks)

->> Calculation of Redundant Bits (r):

Given: $m = 7$ (number of data bits in ASCII)

Hamming formula:

$$2^r \geq m + r + 1$$

Test values of r :

- $r = 3: 2^3 = 8, m + r + 1 = 7 + 3 + 1 = 11 \rightarrow 8 < 11$ ❌
- $r = 4: 2^4 = 16, m + r + 1 = 7 + 4 + 1 = 12 \rightarrow 16 \geq 12$ ✅

✅ Minimum redundant bits $r = 4$

Logic Behind the Inequality:

- 2^r represents the total number of distinct bit patterns the redundant bits can encode.
- $m + r + 1$ accounts for the m data bits, r redundant bits, and 1 for error detection.
- Ensures that all single-bit errors in the codeword can be uniquely detected and corrected.
- This is based on the Hamming distance principle, where a minimum distance of 3 between valid codewords allows single-bit error correction.

Lecture 19

1. Explain why network addresses are considered "logical" and software-based, while MAC addresses are "physical" and hardware-based. Discuss how these two addressing layers interact in the process of Layer-2 and Layer-3 communication. (4 marks)

->> ♦ Logical (IP) Address

- IP address is called logical address because it is assigned by software (OS or network admin).
- It can be changed easily.
- Used at Layer 3 (Network Layer).

- Identifies a device across different networks.
-

◇ Physical (MAC) Address

- MAC address is called physical address because it is permanently assigned to the network interface card (NIC) by the manufacturer.
 - It is hardware-based and usually cannot be changed.
 - Used at Layer 2 (Data Link Layer).
 - Identifies a device within a local network.
-

◇ Interaction Between Layer 2 and Layer 3

When a device sends data:

1. IP address determines the destination network (Layer 3).
2. MAC address is used to deliver the frame within the local network (Layer 2).
3. ARP (Address Resolution Protocol) converts IP address into MAC address.

Example:

If Host A sends data to Host B, it uses B's IP to route the packet and B's MAC to deliver the frame locally.

Thus, IP provides logical routing between networks, while MAC ensures physical delivery within a network.

2. Describe the hierarchical nature of IP addressing and its significance in routing. How does this hierarchy help routers efficiently forward packets to destinations outside a host's subnet? (4 marks)

->> ◇ Hierarchical Nature of IP Addressing

IP addressing is hierarchical because it is divided into two parts:

1. Network ID – Identifies the network.
2. Host ID – Identifies a specific device within that network.

Example:

In IP 192.168.1.10/24

- Network part → 192.168.1.0
- Host part → 10

This structure creates multiple levels (network → subnet → host).

◇ Significance in Routing

- Routers forward packets based only on the network portion, not the full IP address.
 - This reduces routing table size.
 - Supports route aggregation (CIDR).
-

◇ Forwarding Outside the Subnet

When a host wants to send data outside its subnet:

1. It checks the subnet mask.
2. If destination is not in same network, it sends packet to the default gateway (router).
3. Router examines the network ID and forwards it to the correct next network.

Thus, hierarchical addressing allows routers to make efficient forwarding decisions without checking every individual host.

Therefore, IP hierarchy improves scalability and routing efficiency.

3. Compare and contrast Unicast, Multicast, Broadcast, and Anycast addressing modes in terms of their purpose, packet delivery mechanism, and typical use cases in modern networks. (4 marks)

->>

◇ 1. Unicast

- Purpose: One-to-one communication.
- Delivery: One sender → One specific receiver.

- Use Case: Web browsing, email, file transfer.
-

◇ 2. Broadcast

- Purpose: One-to-all communication within a network.
 - Delivery: One sender → All devices in same subnet.
 - Use Case: ARP requests, DHCP discovery.
-

◇ 3. Multicast

- Purpose: One-to-many (selected group).
 - Delivery: One sender → Multiple subscribed receivers.
 - Use Case: Video conferencing, online streaming.
-

◇ 4. Anycast

- Purpose: One-to-nearest (best) receiver.
 - Delivery: One sender → Nearest device among multiple servers with same IP.
 - Use Case: DNS servers, content delivery networks (CDN).
-

☒ Conclusion

- Unicast: One-to-one
- Broadcast: One-to-all
- Multicast: One-to-group
- Anycast: One-to-nearest

Each addressing mode supports different communication needs in modern networks.

4. Explain the role of DNS in network addressing. Describe the process through which a host obtains the IP address of a remote host using DNS and how this address is used in subsequent packet forwarding. (4 marks)

->> ◇ Role of DNS

- DNS (Domain Name System) translates human-readable domain names (like `www.example.com`) into IP addresses.
 - Allows users to access resources without remembering numeric IPs.
-

◇ Process of Obtaining an IP Address

1. Host Request: Host queries its local DNS resolver with the domain name.
2. Recursive Query: Resolver checks its cache or forwards the request to root → TLD → authoritative DNS server.
3. Response: Resolver returns the IP address to the host.

Example: `www.example.com` → `93.184.216.34`

◇ Packet Forwarding Using IP

- Once the host has the IP address, it uses it as the destination in the IP header.
- The packet is forwarded through routers using the network portion of the IP address until it reaches the remote host.

DNS simplifies network addressing and enables proper routing by providing the correct IP for packet delivery.

5. Analyze the routing process in a typical IP network. Discuss how routers use routing tables to determine the next hop, and why routers do not forward broadcast traffic by default. (4 marks)

->>

◇ Routing Process in IP Networks

- Routers connect different networks and forward packets toward their destination.

- Each router uses a routing table containing:

- Destination network
- Next-hop IP
- Subnet mask
- Interface

- When a packet arrives:

1. Router checks destination IP.
2. Matches it with the most specific entry in the routing table.
3. Forwards the packet to the next hop toward the destination.

◇ Why Routers Don't Forward Broadcasts

- Broadcast traffic is meant only for the local subnet.
- Forwarding it would create network congestion and loops.
- Routers isolate broadcast domains to maintain network efficiency and security.

Thus, routers use routing tables for efficient packet delivery and prevent broadcast storms by not forwarding broadcast traffic.

1. Explain the process of Address Resolution Protocol (ARP) and its role in enabling communication between devices on the same broadcast domain. Discuss why ARP caching is important for network efficiency.

->> ◇ Role of ARP

- ARP (Address Resolution Protocol) maps a device's IP address (Layer 3) to its MAC address (Layer 2).
- Essential for devices on the same broadcast domain to communicate using Ethernet frames.

◇ ARP Process

1. Host wants to send data to a known IP but unknown MAC.
 2. Sends ARP request as a broadcast: "Who has IP X.X.X.X?"
 3. Host with that IP replies with its MAC address.
 4. Sender can now deliver frames directly to the recipient.
-

◇ Importance of ARP Caching

- Stores IP-to-MAC mappings temporarily.
- Reduces repeated broadcasts.
- Improves network efficiency and decreases latency.

ARP enables devices to communicate on the same subnet efficiently, and caching optimizes this process.

2. Describe the function of Internet Control Message Protocol (ICMP) in network diagnostics and error reporting. Explain how ICMP-echo and ICMP-echo-reply are used to test network connectivity and diagnose problems.

->>

◇ Function of ICMP

- ICMP (Internet Control Message Protocol) works at the network layer.
 - Used for:
 - Error reporting (e.g., Destination Unreachable, Time Exceeded)
 - Network diagnostics (testing connectivity, path tracing)
-

◇ ICMP Echo & Echo Reply

- ICMP Echo (Ping request): Sent by a host to test reachability of another host.
- ICMP Echo Reply (Ping reply): Response from the destination host if reachable.

Example: Using ping command:

- Host A sends Echo → Host B responds with Echo Reply → Confirms connectivity.

- If no reply → Indicates network issues like unreachable host or dropped packets.

ICMP helps detect network problems and verify end-to-end connectivity.

Lecture 24

1. Explain the role of TCP flags (SYN, ACK, FIN, RST, PSH, URG) in connection management, data transfer, and error handling. Illustrate how these flags are used in a typical TCP three way handshake and connection termination. (4 marks)

->> TCP Flags and Their Roles:

- SYN (Synchronize): Initiates a connection between sender and receiver.
- ACK (Acknowledgment): Confirms receipt of data or control messages.
- FIN (Finish): Signals the sender has finished sending data and wants to close the connection.
- RST (Reset): Terminates a connection abruptly in case of errors.
- PSH (Push): Tells the receiver to deliver data immediately to the application.
- URG (Urgent): Marks data as high-priority for immediate processing.

Three-Way Handshake (Connection Setup):

1. SYN: Client → Server to request a connection.
2. SYN + ACK: Server → Client to acknowledge and synchronize.
3. ACK: Client → Server to confirm the connection is established.

Connection Termination:

- FIN → ACK → FIN → ACK sequence ensures both sides close the connection gracefully.

Conclusion:

TCP flags manage connection setup, reliable data transfer, and orderly connection termination, ensuring robust communication.

2. Describe TCP's congestion control mechanism, including Slow Start, Additive Increase/Multiplicative Decrease, and Timeout React. Explain how the window size is adjusted in response to network conditions and packet loss. (4 marks)

->> TCP Congestion Control Mechanism:

1. Slow Start:

- At the start of a connection, TCP sets the congestion window (cwnd) to a small value and doubles it each round-trip time (RTT) to quickly probe available bandwidth.

2. Additive Increase / Multiplicative Decrease (AIMD):

- Additive Increase: cwnd is increased linearly for each successful RTT to gradually use more bandwidth.
- Multiplicative Decrease: On detecting packet loss, cwnd is halved to reduce congestion.

3. Timeout React:

- If an ACK is not received within a timeout, TCP assumes severe congestion and resets cwnd to initial size, restarting Slow Start.

Window Size Adjustment:

- TCP adjusts cwnd based on network feedback:
 - Successful transmission → increase window (more throughput)
 - Packet loss → decrease window (reduce congestion)

Conclusion:

This dynamic adjustment ensures efficient utilization of network bandwidth while preventing congestion collapse.

3. Analyze how TCP ensures reliable data delivery through sequence numbers, acknowledgements, and retransmission timers. Discuss what happens when segments arrive out of order or are lost, and how duplicate segments are handled. (4 marks)

->> TCP Reliable Data Delivery:

- **Sequence Numbers:** Each byte of data is numbered to keep track of order.
- **Acknowledgements (ACKs):** Receiver sends ACKs for successfully received segments.
- **Retransmission Timers:** If an ACK is not received within a timeout, the sender retransmits the segment.

Handling Out-of-Order or Lost Segments:

- Out-of-order segments are **buffered** until missing segments arrive.
- Lost segments trigger **retransmission** based on timeouts or duplicate ACKs.

Handling Duplicate Segments:

- TCP detects duplicates using **sequence numbers** and **discards repeated segments**, ensuring no data is delivered twice.

Conclusion:

By combining sequence numbers, ACKs, and timers, TCP guarantees **reliable, ordered, and loss-free delivery** over unreliable networks.

4. Compare the functions of the four TCP timers: Keep-alive, Retransmission, Persist, and Timed-Wait. Explain how each timer contributes to connection stability, flow control, and crash recovery. (4 marks)

->> TCP Timers and Their Functions:

1. Keep-Alive Timer:

- Sends periodic probes on idle connections to check if the peer is still active.
- Ensures **connection stability** by detecting dead peers.

2. Retransmission Timer:

- Retransmits segments if an **ACK is not received within the timeout**.
- Maintains **reliable delivery** and handles packet loss.

3. Persist Timer:

- Prevents **deadlock in flow control** when the receiver's window is zero by periodically probing the window status.
- Ensures **continuous communication**.

4. Timed-Wait Timer:

- Keeps the connection in the **TIME_WAIT state** after closure to ensure all delayed packets are received.
- Provides **safe connection termination** and avoids old duplicate segments interfering with new connections.

Conclusion:

Together, these timers manage **reliability, flow control, and connection stability**, and help recover from errors or crashes.

5. Discuss how TCP multiplexing works and its impact on server performance. Explain the benefits and potential drawbacks of keeping TCP sessions open for longer periods, especially in the context of HTTP/1.1 and modern web applications. (4 marks)

->> TCP Multiplexing and Server Performance:

- TCP multiplexing allows **multiple applications or clients** to use the same IP address by differentiating sessions with **port numbers**.
- This enables a server to **handle multiple simultaneous connections** efficiently, improving utilization of network resources.

Keeping TCP Sessions Open:

- **Benefits:** Reduces the overhead of repeatedly establishing connections (handshake), improving performance for protocols like **HTTP/1.1 persistent connections**.
- **Drawbacks:** Long-lived sessions consume **server memory and resources**, which can lead to **resource exhaustion** under heavy load.

Conclusion:

TCP multiplexing enables concurrent communication, while persistent connections **improve performance but require careful resource management** in modern web applications.

1. Compare and contrast UDP and TCP in terms of reliability, connection management, header complexity, and suitability for different types of applications. Provide examples where UDP is preferred over TCP. (4 marks)

->> UDP vs TCP Comparison:

Feature	TCP	UDP
Reliability	Reliable; uses ACKs, sequence numbers, and retransmissions	Unreliable; no guarantees of delivery
Connection Management	Connection-oriented; requires a three-way handshake	Connectionless; no handshake required

Feature	TCP	UDP
Header Complexity	Larger header (20 bytes) with control fields	Smaller header (8 bytes); simpler
Suitability	Applications needing reliable, ordered delivery (e.g., web browsing, email, file transfer)	Applications tolerating loss or needing low latency (e.g., DNS, streaming, online gaming)

Examples Where UDP Is Preferred:

- **DNS queries** – fast, low-overhead request-response.
- **Live video/audio streaming** – tolerates some loss to reduce delay.
- **Online gaming** – low latency is critical, occasional packet loss is acceptable.

Conclusion:

TCP ensures **reliability and order**, while UDP provides **speed and low overhead** for real-time or lightweight applications.

2. Explain why UDP is considered an “unreliable” protocol and discuss the implications of this unreliability in real-time applications such as VoIP and video streaming. How does the underlying IP protocol’s “best-effort” delivery mitigate some of these issues? (4 marks)

->> Why UDP Is “Unreliable”:

- UDP does **not provide acknowledgments, sequencing, or retransmission**, so there is **no guarantee that packets will arrive, arrive in order, or arrive only once**.

Implications for Real-Time Applications:

- Applications like **VoIP and video streaming** may experience **packet loss or out-of-order delivery**, but these apps can tolerate some loss to maintain **low latency**.
- Delaying data for retransmission would harm **real-time performance**.

Role of IP’s Best-Effort Delivery:

- The underlying **IP protocol forwards packets without guarantees**, complementing UDP’s lightweight nature.
- Best-effort delivery ensures **packets are sent as efficiently as possible**, allowing real-time applications to continue functioning despite some losses.

Conclusion:

UDP's unreliability is acceptable for **time-sensitive applications** where speed and low delay are more critical than perfect accuracy.

3. Analyze the UDP header structure and explain the purpose of each field (Source Port, Destination Port, Length, Checksum). Why is the checksum optional in IPv4, and what are the security or reliability risks of disabling it? (4 marks)

->> UDP Header Structure and Field Purposes:

1. **Source Port (16 bits):** Identifies the sending application's port.
2. **Destination Port (16 bits):** Identifies the receiving application's port.
3. **Length (16 bits):** Specifies the total length of the UDP header + data.
4. **Checksum (16 bits):** Used to detect errors in the header and data; ensures **data integrity**.

Why Checksum is Optional in IPv4:

- IPv4 provides a **best-effort delivery**, and the checksum is optional to **reduce processing overhead**.

Risks of Disabling Checksum:

- **Data corruption may go undetected**, leading to application errors.
- **Security risk:** malicious tampering with packets might not be detected.
- Reliability of data delivery is compromised.

Conclusion:

The UDP header is **simple and lightweight**, but disabling the checksum increases the risk of **undetected errors**.

4. Discuss the role of UDP in query-based communications and protocols like DNS and SNMP. Why is UDP's stateless and connectionless nature advantageous in these contexts? (4 marks)

->> Role of UDP in Query-Based Communications:

- UDP is widely used in **query-response protocols** like **DNS** (Domain Name System) and **SNMP** (Simple Network Management Protocol).

- The client sends a **short query**, and the server responds; **no connection setup** is needed.

Advantages of UDP's Stateless, Connectionless Nature:

1. **Low Overhead:** No three-way handshake, so queries are processed **faster**.
2. **Scalability:** Servers can handle **many simultaneous requests** without maintaining per-client state.
3. **Efficiency:** Ideal for **small, short-lived transactions** where reliability is handled at the application level if needed.

Conclusion:

UDP's simplicity, low latency, and statelessness make it **perfect for fast, lightweight query-response protocols** like DNS and SNMP.

5. Evaluate the trade-offs between using UDP and TCP for multimedia streaming. Consider factors such as latency, bandwidth efficiency, packet loss tolerance, and congestion control. (4 marks)

->> UDP vs TCP for Multimedia Streaming:

UDP:

- **Latency:** Very low, since no connection setup or retransmission delays.
- **Bandwidth Efficiency:** High, due to minimal header overhead and no retransmissions.
- **Packet Loss Tolerance:** Acceptable, as minor loss in audio/video can be tolerated.
- **Congestion Control:** None, so streams can continue uninterrupted but may contribute to network congestion.

TCP:

- **Latency:** Higher, due to three-way handshake and retransmissions.
- **Bandwidth Efficiency:** Lower, due to acknowledgments and retransmissions.
- **Packet Loss Tolerance:** Low, retransmissions introduce delays, disrupting real-time playback.
- **Congestion Control:** Ensures fairness but may cause buffering and stalls.

Conclusion:

UDP is preferred for **live audio/video streaming** where **low latency is critical**, while TCP is

better for **file-based or buffered media** where reliability is more important than real-time delivery.

Lecture 28

1. Compare and contrast DNS and HTTP in terms of their transport layer protocols (UDP vs. TCP), communication models, and roles in the Internet infrastructure. Explain why DNS typically uses UDP despite being a critical service. (4 marks)

->>

Comparison of DNS and HTTP:

Feature	DNS	HTTP
Transport Protocol	Primarily UDP (sometimes TCP for large responses)	TCP (requires reliable connection)
Communication Model	Query-Response (request sent, reply received)	Client-Server request-response with persistent connections
Role in Internet	Resolves domain names to IP addresses , enabling connectivity	Transfers web content (HTML, images, etc.) between server and client

Why DNS Uses UDP:

- Most queries and responses are **small**, so UDP's low overhead improves speed.
- DNS is **time-sensitive**, and avoiding the TCP handshake reduces latency.
- Reliability is **handled at the application level**, so UDP is sufficient for most lookups.

Conclusion:

DNS prioritizes **speed and low overhead**, while HTTP prioritizes **reliability and ordered data transfer**.

2. Explain the process of email transmission using SMTP, POP3, and IMAP. Discuss the roles of User Agents, Message Transfer Agents, and mail servers, and analyze the differences between POP3's delete and keep modes. (4 marks)

->> Email Transmission Process:

1. **User Agent (UA):** Software like Outlook or Gmail used by the user to **compose, send, and read emails**.
2. **Message Transfer Agent (MTA):** Server software (e.g., Postfix, Exchange) that **forwards emails** from sender to recipient using **SMTP**.
3. **Mail Servers and Protocols:**
 - **SMTP:** Transfers outgoing mail from the UA to the MTA and between MTAs.
 - **POP3 (Post Office Protocol 3):** Retrieves emails from the server.
 - **Delete mode:** Emails are removed from the server after download.
 - **Keep mode:** Emails remain on the server for access from multiple devices.
 - **IMAP (Internet Message Access Protocol):** Synchronizes emails across devices, keeping messages **on the server** for consistent access.

Conclusion:

SMTP handles **sending**, while POP3 and IMAP handle **retrieving emails**. POP3 is simpler but less flexible; IMAP is preferred for **multi-device access**.

3. Describe how FTP uses out-of-band control and data connections (ports 20 and 21). Discuss the advantages and disadvantages of this approach compared to using a single connection for both control and data, as in HTTP. (4 marks)

->> FTP Out-of-Band Control and Data Connections:

- FTP uses **two separate TCP connections**:
 - **Port 21 (Control Connection):** Sends commands and responses (e.g., login, directory listing).
 - **Port 20 (Data Connection):** Transfers actual files between client and server.

Advantages:

- **Separation of control and data** allows commands to be sent while data is being transferred.
- **Efficient management** of multiple data transfers without interrupting the control channel.

Disadvantages:

- **Complexity:** Requires managing two connections per session.
- **Firewall/NAT issues:** Easier to block FTP data port than HTTP's single port.

Comparison with HTTP:

- HTTP uses a **single connection** for both control (requests) and data (responses), making it **simpler and more firewall-friendly** but less flexible for simultaneous commands and data transfers.

Conclusion:

FTP's dual connections provide **flexibility and parallelism**, while HTTP's single connection prioritizes **simplicity and ease of access**.

4. Analyze the evolution of HTTP from version 1.0 to 1.1 in terms of connection persistence. Explain the impact of persistent and non-persistent connections on web performance, server load, and latency. (4 marks)

->> Evolution of HTTP Connection Management:

- **HTTP/1.0:** Uses **non-persistent connections**; each request/response pair requires a **new TCP connection**, leading to **higher latency** and increased server load.
- **HTTP/1.1:** Introduces **persistent connections** by default; multiple requests and responses can **share the same TCP connection**, reducing the overhead of repeated handshakes.

Impact on Web Performance:

- **Persistent connections:** Lower latency, better bandwidth utilization, and reduced CPU/memory load on servers.
- **Non-persistent connections:** Higher latency due to frequent connection setup/teardown, more overhead on server, and slower page load times.

Conclusion:

HTTP/1.1's persistent connections significantly **improve web performance and scalability** compared to HTTP/1.0.

5. Explain why HTTP is considered a stateless protocol and discuss the mechanisms (such as cookies and sessions) used to maintain state in web applications. How does statelessness affect scalability and security? (4 marks)

->> HTTP as a Stateless Protocol:

- HTTP is **stateless**, meaning each request from a client is **independent** and the server does not retain information about previous requests.

Mechanisms to Maintain State:

1. **Cookies:** Small pieces of data stored on the client to track user information across requests.
2. **Sessions:** Server-side storage linked to a session ID, allowing the server to remember user-specific data during interactions.

Effects of Statelessness:

- **Scalability:** Easier to scale servers, as no session data needs to be maintained for each client by default.
- **Security:** Requires careful handling of cookies and session IDs to prevent **session hijacking** or unauthorized access.

Conclusion:

Statelessness simplifies server design and improves scalability, but web applications rely on cookies and sessions to provide a **stateful user experience** while managing security risks.